Workflow 1: Biomanufacturing of Styrene

DESIGN

- **Chassis selection and metabolic model retrieval:** Select an appropriate host and retrieve its genome-scale metabolic model. (BiGG Models)
- **Metabolic pathway prediction**: Identify possible biosynthetic routes from endogenous precursors to styrene. (RetroPath2.0)
- **Pathway enumeration and reconstruction:** Enumerate the reactions constituting each predicted pathway and reconstruct complete candidate pathways. (RP2Paths, rpCompletion)
- **Pathway feasibility and ranking:** Evaluate predicted routes using FBA, thermodynamics and pathway scoring. (rpFBA, rpThermo, rpRank, COBRApy)
- **Enzyme identification and selection:** Identify enzymes and suitable homologs for the proposed reactions. (Selenzyme, EnzymeMiner)
- **Metabolic engineering strategy:** Identify gene deletions or overexpression targets to increase precursor supply and styrene production. (OptFlux)
- **Genetic regulatory element selection:** Design or select promoters, RBSs and other regulatory elements for pathway expression. (PartsGenie, RBS Calculator)
- **Combinatorial experimental design**: Select promising combinations of enzymes, regulatory parts and genetic modifications. (OptDoE)
- **DNA construct and assembly design:** Design the DNA constructs and generate an assembly strategy. (DNA Weaver, rpBASICDesign, AssemblX)
- **Cultivation experimental design:** Define cultivation conditions, experimental factors and biological replicates for evaluating the candidate strains. (OptDoE)

BUILD

- **Automated DNA assembly protocol generation**: Translate the genetic designs into instructions for automated DNA assembly using a liquid-handling robot. (DNA-BOT)
- **Genetic construct verification**: Compare sequencing results against the expected DNA sequences to verify the assembled constructs. (TraceTrack)
- **Strain and sample tracking:** Register the constructed strains, their genetic modifications and their associated experimental samples. (eLabFTW)

TEST

- **Microbial growth analysis:** Process growth curves to quantify growth rates and assess the effects of styrene toxicity. (omniplate)
- **Styrene and substrate quantification:** Process chromatographic measurements and calculate product titer, yield and productivity. (OpenChrom, R/Python for derived metrics)

LEARN

- **Metabolic bottleneck identification:** Compare experimental results with metabolic predictions and investigate limitations in precursor availability or pathway flux. (CellNetAnalyzer, COBRApy)
- **Predictive modelling and next-iteration design:** Train regression models using genetic designs, cultivation conditions and experimental production data. Predict the performance of untested strain designs and identify promising candidates for the next DBTL iteration (scikit-learn, OptDoE)